Google Search in the Age of AI Agents

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Personal analysis, not investment advice. Written with the assistance of Claude (Anthropic).

How Google earns from search today

Google Search earns mainly by auctioning ads beside results and charging the advertiser each time a human clicks one. In Q3 2026 (Jul–Sep), Search & other brought in $63.35B, about half of Alphabet's $127.6B revenue.

$$\text{Ad revenue} \approx \text{Paid clicks} \times \text{CPC}$$

Paid clicks = number of times users click on ads; CPC (cost per click) = average price advertisers pay per click, set in Google's auctions.

- **Who pays:** the merchant or advertiser. Commercial queries (shopping, travel, insurance, finance) carry most of the revenue; most informational queries earn nothing.
- **Organic clicks** earn nothing directly, but the traffic keeps merchants dependent on Google and willing to advertise.
- **Shopping ads** draw on the Shopping Graph: 60B+ product listings fed mainly by Merchant Center, about 2B refreshed per hour.
- **Rough average:** about $0.05 per query (Q3 revenue ÷ ~1.25T queries; approximate, query count from Google's older public figure).
- **Recent trend:** Search growth slowed from 19% (Q1 2026) to 17% (Q2) and 12% (Q3). Google attributed part of the slowdown to AI Mode, which monetizes at a lower rate than classic search.

What replaces the click model with agents

When software instead of a person reads the results, there are no ad views and no ad clicks: agent clicks count as invalid traffic and advertisers don't pay for them. Revenue has to come from other models, all of which already exist in some form.

Stage	Who reads the results	How it is monetized	Status (Oct 2026)
Classic search	Human	Ads, paid per click by the merchant	Core business, slowing
AI Mode / AI Overviews	Human	Ads, but fewer slots and fewer clicks	Live; lower monetization confirmed by Google
Agent search via API	Machine	Per-query API fee, paid by the agent operator	Live: Gemini API grounding, Brave, Exa, Tavily

Stage	Who reads the results	How it is monetized	Status (Oct 2026)
Agentic checkout	Machine, for a human	Fee or commission per completed purchase, paid by the merchant	Live: Google UCP, ChatGPT (ACP), Copilot Checkout, Perplexity

The per-conversion model keeps the merchant as payer, which made ads so profitable. But a search engine paid only on purchases carries the cost of every query that doesn't convert:

$$\text{Cost per paid conversion} = \frac{c}{p}$$

c = the engine's cost to serve one query; p = probability that a query ends in a purchase credited to that engine. Per query, costs are tiny (a fraction of a cent), but when p is small they approach the fee itself. Engines therefore push for per-query fees, while subscription-funded agents push for cheap or free queries.

E-commerce platforms and why they threaten Google

E-commerce platforms (also called commerce platforms or, in German, Shopsysteme) are software services that run a merchant's online shop: storefront, product catalog, inventory, checkout and payments. Shopify is the largest independent one; others include BigCommerce, WooCommerce, Shopware, Adobe Commerce and Salesforce Commerce Cloud.

- **How they earn:** subscription fees plus a cut of every sale through payment processing. As far as I know, most of Shopify's revenue comes from these payment-related merchant solutions. They are paid per transaction, not per click.
- **What changed:** they now publish merchants' catalogs and checkout to AI agents. Shopify's Agentic Storefronts, on by default for eligible stores since late March 2026, send products to ChatGPT, Google AI Mode, Gemini, Microsoft Copilot and Perplexity. Feed management tools (Feedonomics, Feedoptimise, Pricefy) do the same for merchants on other systems.

Google offers comparable services (Merchant Center, the Shopping Graph with 60B+ listings, agentic checkout via UCP), yet these platforms still weaken its position:

1. **Google's listings came from its traffic, not its tools.** Merchants fed Google because the shoppers were there. Once agents bring the shoppers, the incentive to feed Google first disappears.
2. **Portability removes exclusivity.** One toggle sends the same catalog to every agent, so rivals get most of what Google's graph holds for those merchants. Google's larger listing count then matters mainly for the long tail outside these platforms.
3. **Neutral position vs. a bet.** A platform earns on every sale whichever agent the shopper uses. Google earns only if its own surfaces (Search, AI Mode, Gemini) win the shopper.
4. **The merchant relationship moves to the platform.** The platform holds the catalog, checkout and payment data; Google becomes one sales channel among many, and channels can be switched off.
5. **Different business model.** Google's shopping services are a funnel for ads: free listings attract merchants, paid ads earn the money. Platforms monetize the transaction itself, which is the model agentic commerce needs. To follow, Google must compete with them for the same transaction fee.
6. **Bargaining power shifts.** Whoever controls access to many merchants' catalogs can set terms for agents. That used to be Google's role.

Groups of common interest

The agent shift splits the market into players whose advantage came from buying or controlling attention, and players who gain when agents compare offers neutrally. These are overlapping interests, not formal alliances: everyone is hedging by joining every agent platform.

Group	Members	Common interest	Where it breaks down
Attention incumbents	Google, large merchants (Walmart, Target, Otto, Zalando)	Visibility bought through ad budgets and brand recognition; both lose when an agent ranks offers on price, reviews and delivery alone	Large merchants want to cut their ad spend and feed every agent directly (Walmart supports both ACP and UCP); Amazon competes with Google in product search
Agent economy	Agent providers (OpenAI, Microsoft Copilot, Perplexity), e-commerce platforms (Shopify, BigCommerce, Shopware), payment providers (Stripe, PayPal, Visa, Mastercard), small and mid-size merchants	Neutral comparison levels the field for small merchants; platforms and payments earn per transaction whichever agent wins; agents want cheap data and checkout	Agents and platforms compete for the transaction fee; a dominant agent could bypass platforms or squeeze their fees

Why the incumbents' interests overlap: in Google's ad auctions, large merchants can outbid smaller ones for top placement. Agents remove that advantage, threatening both Google's ad revenue and the large merchants' budget and brand edge.

Why it is not an alliance: Google also cooperates with the agent economy. Shopify feeds its merchants into AI Mode and Gemini by default, Google's checkout standard (UCP) is shared with Microsoft, and Google needs small merchants to keep the breadth of its Shopping Graph. Large merchants, for their part, connect to every agent rather than relying on Google.

Challenges for Google

The core problem: each step from classic search to AI Mode to agents moves traffic away from the click model that produces about half of Alphabet's revenue.

1. **Lower monetization per query.** AI answers have fewer ad slots and send fewer clicks; agents generate no ad revenue at all.
2. **Ranking matters less.** Agents can read hundreds of results and sort them themselves, eroding Google's decades-long lead in ranking for humans.
3. **Competition shifts to data.** What counts becomes index coverage and freshness, especially real-time product data. Google still leads (60B+ listings), but the gap is narrower than its ad dominance.
4. **Merchant data is becoming portable.** E-commerce platforms such as Shopify, plus feed management tools, send one catalog to every agent (ChatGPT, Gemini, Copilot, Perplexity), so being in Google's Shopping Graph stops being exclusive.
5. **Commoditization of search supply.** Agents switch providers with one line of code and can query several engines cheaply, pushing API prices toward cost.
6. **Value moves to both ends of the chain:** the agent the user talks to, and the merchant platforms and payment providers that complete the sale. Google's middle position, routing shoppers to merchants, is most at risk.

7. **The worst case:** small merchants reach agents via e-commerce platforms, large merchants list directly with their chosen agents, and Gemini fails to become a major agent. Google is then bypassed in shopping entirely.
8. **Funding pressure.** Capex is above $200B in 2026 and expected above $300B in 2027; Q3 free cash flow was −$9.1B, and long-term debt rose from $46.5B to $125.1B in nine months. Weaker search cash flow makes this harder to carry.
9. **Depreciation risk.** Servers are depreciated over six years, a figure set before AI accelerators dominated capex. With performance per watt roughly doubling every 1–2 years, chips may become uneconomic before six years where power is the constraint, and value loss is front-loaded; see the chapter on chip generations below.

Google's hedges and what to watch

Alphabet holds a position in almost every layer of the AI stack, so even the worst case shrinks it rather than removing it from the loop.

- **Owning the agent:** Gemini, distributed through Android, Chrome, Search and Workspace (Gemini app above 1B users).
- **Supplying the data:** the web index and the Shopping Graph, sold to others via APIs if Gemini loses.
- **Supplying the infrastructure:** Cloud and TPUs earn whichever agent wins. Cloud revenue grew 103% to $30.7B in Q3 2026, about a quarter of Alphabet's revenue.
- **Unaffected businesses:** YouTube, Android and Play, and subscriptions earn largely independently of search.

Metrics to watch each quarter:

- Search revenue growth against query growth
- Gemini users against ChatGPT's
- Cloud operating income against depreciation, plus Cloud backlog
- Free cash flow and any change to server useful lives in the 10-Q

Chip generations and useful life

AI chips lose economic value much faster than six years suggests: at equal precision, peak efficiency has grown about 1.5× a year. Whether that forces early replacement depends on which resource is scarce: power or chips.

How efficiency has evolved

Efficiency is measured as peak floating-point throughput per watt:

$$\text{Perf/W} = \frac{\text{Peak FLOPS}}{\text{TDP}}$$

Peak FLOPS = the chip's theoretical maximum floating-point operations per second at a given precision, without sparsity; TDP (thermal design power) = the maximum sustained power the chip package is rated for, in watts.

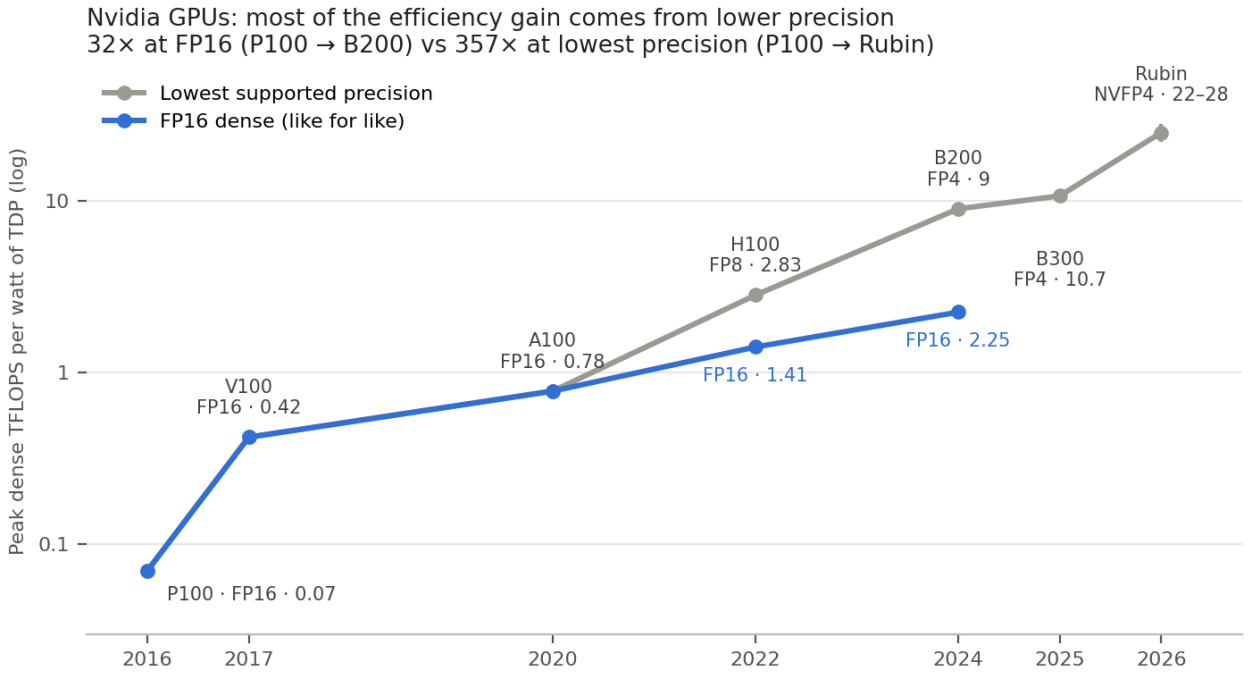


Figure 1: P100–B200: Nvidia datasheets, approximate figures from memory. B300 and Rubin: 2026 reporting; Rubin TDP (1,800–2,300 W) not confirmed by Nvidia.

Google publishes its TPU figures only as multiples of earlier generations, measured in peak FP8 FLOPS per watt of TDP:

TPU generation	Announced	Perf/W vs TPU v2
TPU 8t / 8i	2026	up to ~60× (derived: up to 2× Ironwood)
Ironwood (v7)	2025	~30× (Google)
Trillium (v6e)	2024	~15× (derived: Ironwood is 2× Trillium)
TPU v2	2017	1×

Both lines improved roughly 30× over about eight years at like-for-like precision. Caveats: lower precision (FP8, FP4) inflates headline jumps; real workloads reach only a fraction of peak; TDP excludes cooling, host CPUs and networking; and TDP per chip rose from 300 W to roughly 2,000 W.

Scenario A: power is scarce, chips are available

The data centre's megawatts are fully used, so every watt an old chip draws is a watt a new chip could use. Replace when the extra revenue the new chips earn in the same power slot covers their net cost:

$$(R_{\text{new}} - R_{\text{old}}) \times T \;\geq\; C_{\text{new}} - S_{\text{old}}$$

R_{new} = annual revenue the new chips earn in the power slot; R_{old} = annual revenue the old chips earn in the same slot; T = years the new chips will run; C_{new} = purchase and installation cost of the new chips; S_{old} = resale or redeployment value of the old chips (high if spare power exists

elsewhere). Energy cost cancels out, since the power drawn is the same. Ignoring discounting for simplicity.

With perf/W doubling every 1–2 years, R_{new} can be several times R_{old} after two generations, which is why replacement every second or third generation is plausible here.

Scenario B: chips are scarce, power is available

There is no replacement: every new chip goes into free powered space, in addition to the existing fleet. An old chip is retired only when it costs more to run than it earns. Its revenue is not fixed, though: as new generations add capacity, the market price of compute falls, just as old CPUs became nearly worthless.

$$F_{\text{old}} \times p(t) \;<\; E_{\text{old}} + M_{\text{old}}, \qquad E_{\text{old}} = P \times \text{PUE} \times 8760\,\text{h} \times c$$

F_{old} = the old chip's annual compute output, fixed by its hardware; $p(t)$ = market price per unit of compute in year t, so $F_{\text{old}} \times p(t)$ is the chip's annual revenue; E_{old} = its annual energy cost; M_{old} = other annual running costs (maintenance, space, networking); P = the chip's power draw in kW; PUE (power usage effectiveness) = facility overhead factor for cooling and power delivery; 8,760 h = hours in a year; c = electricity price per kWh.

The price $p(t)$ is increasingly set by the newest chips, which produce compute with far less energy per FLOP. Once it falls to about the old chip's own running cost per unit of compute, the chip is uneconomic even though it still works.

Rough timing: an H100 (P = 0.7 kW, PUE ≈ 1.2, c = \$0.10/kWh) costs about \$740 a year in energy and earns very roughly \$15–20k. With other running costs, retirement comes at about \$2–3k a year, so revenue must fall by about 8×. The time this takes follows from the retirement condition.

Find t where $F_{\text{old}} \times p(t) = E_{\text{old}} + M_{\text{old}}$, assuming the number of FLOP per unit of price rises from year to year by a factor g ($g \approx 1.4$, see below). The price of compute then follows

$$p(t) = p_0 \cdot (g^{-1})^t$$

where p_0 = today's market price per unit of compute, $p(0)$. Inserting this into the retirement condition:

$$F_{\text{old}} \cdot p_0 \cdot (g^{-1})^t = E_{\text{old}} + M_{\text{old}}$$

With $R_0 = F_{\text{old}} \times p_0$ and $R_{\text{min}} = E_{\text{old}} + M_{\text{old}}$ (defined below), this becomes

$$(g^{-1})^t = \frac{R_{\text{min}}}{R_0}$$

and taking the logarithm of both sides gives:

$$t = \frac{\ln(R_0/R_{\text{min}})}{\ln g}$$

t = years until retirement; R_0 = the chip's annual revenue today; R_{min} = the revenue at which it just covers its running costs; g = the yearly factor by which compute gets cheaper. Epoch AI found that FLOP/s per dollar for GPUs used in ML research doubles every 2.07 years (data 2006–2021, fixed precision), so $g \approx 2^{1/2.07} \approx 1.4$, about 29% cheaper per year. With $R_0/R_{\text{min}} \approx 8$, $t \approx \ln 8/\ln 1.4 \approx 6$ years.

Observed rental prices bracket this trend. The A100 fell from $2.40 per hour at launch in 2020 to about $1.40 in 2024, about −13% a year (Lepton AI). The H100 fell from over $8 in mid-2023 to about $2.30–2.60 in mid-2026, about −33% a year (Data Gravity, Lines.com), with a 40% rebound between October 2025 and March 2026 when demand outran supply (Crypto Briefing).

Scenario B therefore gives a finite economic life of about six years on the long-run trend: longer while chips stay scarce, shorter when supply catches up quickly.

Which scenario applies, and what it means for depreciation

Today both apply, depending on site and company: Microsoft has reported chips waiting for power, while others still wait for chips. As chip supply catches up and power stays scarce, Scenario A takes over.

Google set its six-year server life in 2023 from historical fleet data. In Scenario B, the long-run price trend supports roughly that figure. The risks lie elsewhere: in Scenario A the economic life is shorter, and in both scenarios value loss is front-loaded, since a chip earns most in its early, scarce years while straight-line depreciation spreads the cost evenly. On roughly $120B of 2026 server purchases, depreciation over six years is about $20B a year; over four years it would be about $30B, about $10B less operating profit per year for that one vintage.

Sources

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